CCCA Meta-Atlas — Gao et al. 2021 (Breast)

Breast Cancer · 10x · 6,606 malignant cells (6 breast samples; thyroid ATC excluded) → 6,605 after island removal

Do malignant cells recapitulate embryonic programs? Non-branching (linear) manifold — Gate 1 GO · Gate 2 GO · Gate 3 NA

GO / NO-GO GATES

Gate 1 — mixing & continuity: **GO**

Harmony (θ=8) on sample, n_pcs swept 20–40: per-neighbourhood iLISI 1.00→1.59 (of 6 samples), rising then plateauing at θ=8. kNN graph is ONE connected component at every θ (3–10); 1 detached cell removed. Patient-territorial but continuous — not an archipelago → GO.

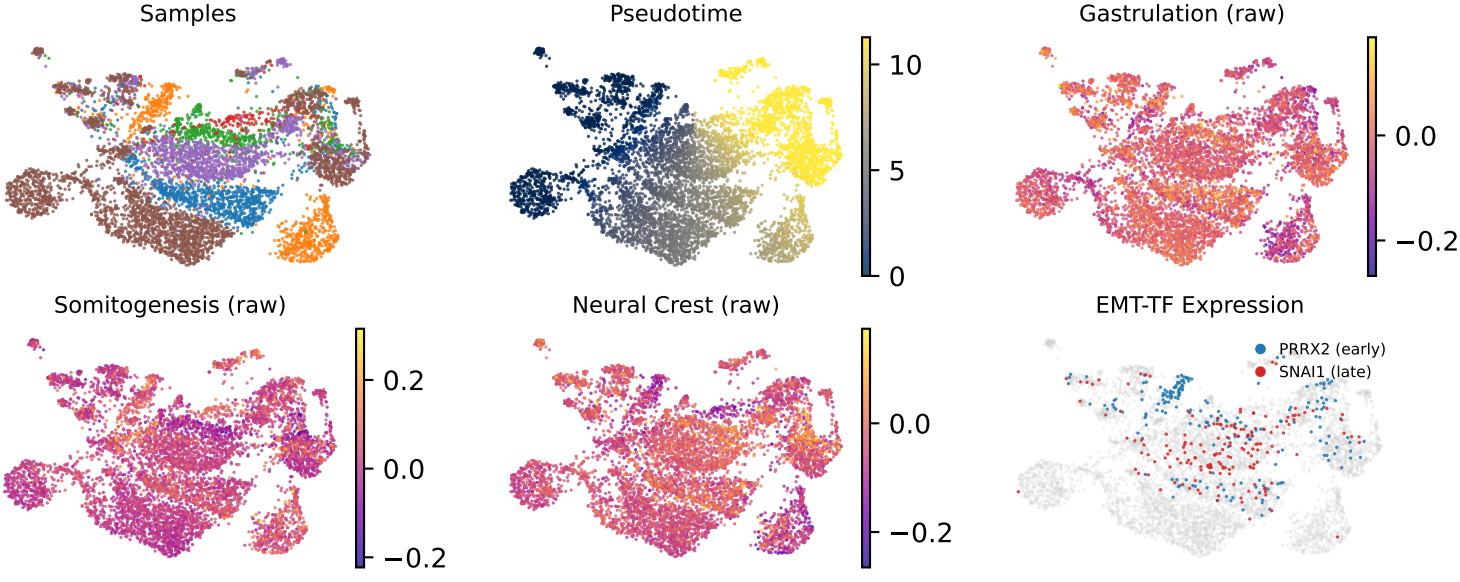
Gate 2 — module geography: **GO (non-branching)**

Smoothed geography: Gast↔pSom anti-correlate (r=−0.50, organized cross-repression); Crest opposes pSom (r=−0.55), weakly tracks Gast (+0.19). Along pseudotime Gast falls (ρ=−0.33) and Crest rises (ρ=+0.24; NC-high decile at t=0.77, terminal). A single Gast/pSom→Crest axis with a terminal Crest gradient — present but not resolved into two competing arms → GO (non-branching case, task §1–2). Crest does not enrich a distinct Gast arm (no arms) — no inversion flag.

Gate 3 — Gast↔pSom cross-repression: **NA**

Non-branching topology. Curve: 2 tips, 0 forks (Nodes 15/20/30), fold-free (self-approach 2.7–2.9). A PPT tree returns 2 forks, but they are reconciled as a node-density artifact, NOT a Gast/pSom Y: (i) all tree arm-segments share one flat profile — Gast−pSom spread across arms 0.009, no Gast-arm vs pSom-arm; (ii) forks dissolve as ppt_sigma rises (2→2→1→0); (iii) the 2 forks sit 2.24 UMAP units apart (median edge 0.84) in the dense centre. No bifurcation → NA.

STEP FIGURES



PARAMETER CHOICES & RATIONALE

Threshold: ≥100 malignant cells/sample → all 6 breast samples kept (smallest TNBC3=195; non-binding). Thyroid ATC excluded (out of scope for a Breast dataset).

Data: Two leaves inner-joined on 22,164 shared genes (different genome annotations; outer join would fabricate ~11k source-confounded zero columns). 2,696 overdispersed genes; no sc.pp.scale before PCA.

Embedding: Harmony θ=8 on sample; n_pcs=30, n_neighbors=30, UMAP spread=1.3.

Trajectory: scFates curve (non-branching), Nodes=20, epg_λ=0.3, epg_μ=100; fold-free (self-approach 2.73; independent diffusion-map check monotone). PPT-tree 2 forks reconciled as a density artifact (see Gate 3).

Root: Gast/pSom-high, Crest-low curve tip (developmental origin: Gast EARLY, Crest LATE). Axis not proliferation-driven (pseudotime vs cycling ρ=+0.40, secondary).

Geometry: Linear — single Gast/pSom→Crest axis. Raw module means: Gast −0.025, pSom +0.020, Crest −0.045.

EMT-TFs: PRRX2 (early/root) ↔ SNAI1 (late/terminus), detection-gated (>5%) from a 10-TF panel incl. FOXC/PRRX families (VIM excluded — mesenchymal marker, not a TF).

Verdict: The embryonic pattern reproduces as a single gastrulation/somitogenesis→neural-crest axis with organized Gast↔pSom cross-repression and a terminal Crest gradient (non-branching); the PPT tree's 2 forks are a density artifact, not a bifurcation, so Gate 3 is NA. Geometry Linear — consistent with the CCCA breast cohort.